

Analisis Data (Non Parametrik)

Menganalisis data primer dan sekunder, mulai dari: Data deskriptif dan non parametric



Time

100 min



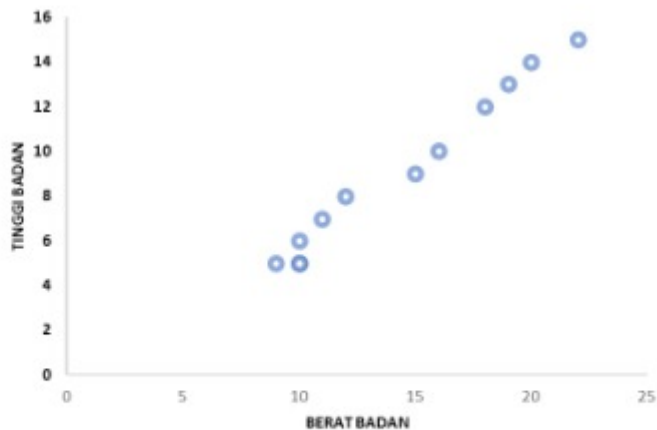
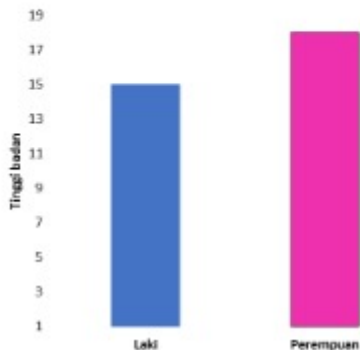
Group

6 - 9



Facilitation experience

focus on group
process



Kategorik

Numerik

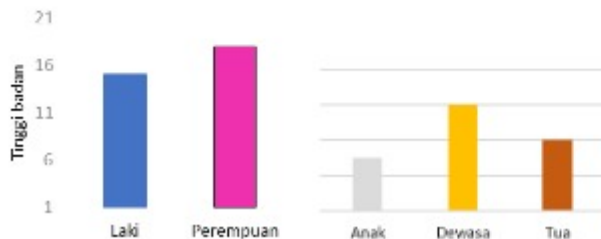
GENDER	KLASIFIKASI UMUR	TINGGI BADAN	BERAT BADAN
Laki	Anak	19	22
Perempuan	Tua	165	41
Perempuan	Dewasa	167	45
Perempuan	Dewasa	166	44
Laki	Dewasa	51	
Laki	Dewasa	48	
Perempuan	Dewasa	42	
Perempuan	Anak	21	20
Laki	Dewasa	177	58
Perempuan	Dewasa	177	57

chi
squared

T test

1 sample
proportion

T-test

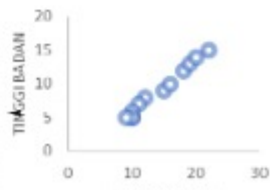
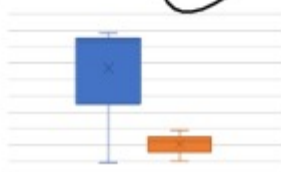


Populasi



Sampel

Temuan



What to do ?

- 👉 **Theory + Business Model** -> Pertanyaan + hypo test
- 👉 **Data Summary** -> Kenali bentuk data
- 👉 Translate data into **insights**
- 👉 Create **data viz, interactive report + dashboard**



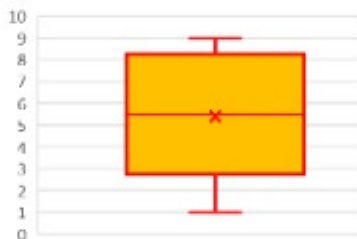


Contoh:

	n	%
Laki	12	60%
Perempuan	8	40%



Min	1
Q1	3,25
Med	5,5
Q3	7,75
Max	9



Column1		
Mean		5,4
Standard Error		0,909212113
Median		5,5
Mode		9
Standard Deviation		2,875181154
Sample Variance		8,266666667
Kurtosis		-1,35698491
Skewness		-0,15146311
Range		8
Minimum		1
Maximum		9
Sum		54
Count		10



Apakah terdapat perbedaan antara jumlah Wanita dan Pria dalam populasi?

H_0 : Tidak ada perbedaan
 H_1 : Ada perbedaan

Test: one sample proportion test (p-value)

$p < \alpha \rightarrow$ tolak H_0 , terima H_1

contoh:

Sig (p-value) = 0.002, artinya kesalahan penelitian secara statistic (uji statistic) sebesar 0.2%

**Pertanyaan
Penelitian &
Hipotesis**

$H_0 : H_1$
Nilai α

Apakah tinggi badan rata-rata berbeda dengan data sebelumnya?

H_0 : Tidak ada perbedaan
 H_1 : Ada perbedaan

Test: T- test (p-value)

$p < \alpha \rightarrow$ tolak H_0 , terima H_1

Apakah ada hubungan antara tinggi badan dan berat badan?

H_0 : Tidak ada hubungan
 H_1 : Ada hubungan

Koefisien korelasi:



Data Insight & Viz

COMPARISON



BAR PLOT

COMPARING CATEGORIES
WITHIN THE SAME MEASURE
OR THE SAME MEASURES.



LINE PLOT

COMPARING TRENDS
OVER TIME.

COMPOSITION



PIE CHART

DISPLAYING A STATIC
COMPOSITION OF VALUES.



WATERFALL CHART

DISPLAYING THE STATIC
COMPOSITION OF A VALUE
WITH ACCUMULATION OR
SUBTRACTION FROM THE TOTAL.

RELATIONSHIP



BOXPLOT

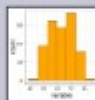
DISPLAYING OUTLIERS
AND DATA CLUSTERS.



SCATTER PLOT

DISPLAYING THE RELATIONSHIP
BETWEEN TWO OR THREE
MEASURES FOR A DIMENSION.

DISTRIBUTION



HISTOGRAM

DISPLAYING THE
DISTRIBUTION OF DATA
INTERVALS.



BOXPLOT

DISPLAYING RANGES AND
DISTRIBUTION OF
NUMERIC DATA.

GEOGRAPHICAL



MAP

DISPLAYING DATA
REPRESENTED
GEOGRAPHICALLY BY
A POINT OR AREA.



Breakout group 1

PART 1

Instructions

1. Discuss the topic below.
2. Create table
3. Define things that you agree upon using sticky-notes.

🕒 15 mins.



Group Presentation...

Other...

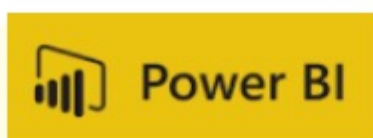


Instructions

1. Prep. device & app (install)
 2. Choose type of data viz
- 🕒 next meeting

PART 2

Where do we work?



Data Sources ?



atlas.cid.harvard.edu
The Atlas of Economic Complexity by @HarvardGrwthLab
Visualize global trade data and economic growth opportunities for every country

 public.opendata...



Airbnb - Listings
Inside Airbnb is an independent, non-commercial set of tools and data that allows you to explore how Airbnb is really being used in cities around the world. By analyzing publicly available information about a city's Airbnb's listings, Inside Airbnb provi...



To Do List

Install
Power
BI



Data



Data
Summary



Visual &
Analysis



Share @
Teams

HOW WE STAY CONNECTED



Congrats! 🎉

You and your team have opened the discussion on the *way of working* together. Well done! Make sure to check-in with each other on regular basis as your team's ideal way of working will continuously change.